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Image capturing apparatus

Abstract

An image capturing apparatus includes: a first heat-dissipating member to dissipate heat generated by a first heat source; at least one second heat-dissipating member to dissipate heat generated by at least one second heat source; a duct inside which the first heat-dissipating member is disposed, and configured to guide air drawn in from an inlet port toward a first outlet port; and a fan disposed between the first outlet port and the first heat-dissipating member inside the duct and configured to discharge air toward the first outlet port from the inlet port. At least one opening is positioned between the fan and first outlet port, part of the air discharged toward the first outlet port by the fan is pushed out from the opening, and the air pushed out from the opening cools the second heat-dissipating member and is discharged from a second outlet port.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to an image capturing apparatus.

Description of the Related Art

(2) Japanese Patent Laid-Open No. 2013-85204 discloses a structure in which a fan is provided at each of an inlet port and an outlet port provided in an exterior portion of an image capturing apparatus, outside air is drawn in from the inlet port, and air heated by each component inside the image capturing apparatus is discharged to the outside by the fan at the outlet port to perform

overall ventilation.

(3) Japanese Patent Laid-Open No. 2020-184771 discloses a structure in which a duct is provided between an inlet port and an outlet port provided in an exterior portion of an image capturing apparatus, outside air is drawn in by a fan provided at an inlet port of the duct, and the air inside the duct is pushed out to the outside to ventilate the inside of the duct.

SUMMARY OF THE INVENTION

(4) An image capturing apparatus whose exterior is provided with at least one inlet port for drawing in air from outside of the image capturing apparatus and a first outlet port and a second outlet port for discharging air from inside of the image capturing apparatus includes: a first heat-dissipating member configured to dissipate heat generated by a first heat source; at least one second heat-dissipating member configured to dissipate heat generated by at least one second heat source; a duct inside which the first heat-dissipating member is disposed, and configured to guide air drawn in from the at least one inlet port toward the first outlet port; and a fan disposed between the first outlet port and the first heat-dissipating member inside the duct and configured to discharge air toward the first outlet port from the at least one inlet port. In the image capturing apparatus, at least one opening is provided in the duct, the at least one opening is positioned between the fan and the first outlet port, part of the air discharged toward the first outlet port by the fan is pushed out from the at least one opening, and the air pushed out from the at least one opening cools the at least one second heat-dissipating member and is discharged from the second outlet port.

(5) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of an image capturing apparatus according to a first exemplary embodiment.

(2) FIG. 2 is a perspective view of the image capturing apparatus according to the first exemplary embodiment.

(3) FIG. 3 is a perspective view of the image capturing apparatus according to the first exemplary embodiment.

(4) FIG. 4 is an exploded view of the image capturing apparatus according to the first exemplary embodiment.

(5) FIG. 5 is a sectional view of the image capturing apparatus according to the first exemplary embodiment.

(6) FIG. 6 is a perspective view of a cooling mechanism of the image capturing apparatus according to the first exemplary embodiment.

(7) FIG. 7 is a rear view of the image capturing apparatus according to the first exemplary embodiment.

(8) FIG. 8 is a sectional view of the image capturing apparatus according to the first exemplary embodiment.

(9) FIG. 9 is a perspective view of an image capturing apparatus according to a second exemplary embodiment.

(10) FIG. 10 is an exploded view of the image capturing apparatus according to the second exemplary embodiment.

DESCRIPTION OF THE EMBODIMENTS

(11) Exemplary embodiments of the present invention are described in detail below with reference to the attached drawings. The exemplary embodiments described below are examples for realizing the present invention, should be corrected or modified as appropriate according to the structure or

various conditions of an apparatus to which the present invention is applied, and the present invention is not intended to be limited to the exemplary embodiments below. Parts of the exemplary embodiments described below may be combined as appropriate.

First Exemplary Embodiment

(12) A box-type lens-interchangeable image capturing apparatus is described below as an image capturing apparatus according to an exemplary embodiment. FIGS. **1**, **2**, and **3** are each a perspective view of an image capturing apparatus **100** according to the present exemplary embodiment.

(13) The image capturing apparatus **100** has a lens mount **111**, and an interchangeable lens can be attached to the lens mount **111**. The lens mount **111** can be electrically connected to the interchangeable lens through a lens connector **112**. Although, in the present exemplary embodiment, a bayonet mount structure is provided, a lens mount of any other form may be used. Instead of a lens-interchangeable image capturing apparatus, an integrated-lens image capturing apparatus may be used.

(14) The image capturing apparatus **100** includes a power input terminal **131**, connectors **132** for sending a signal to and receiving a signal from an external device, various keys **133** used by a user for performing operations and settings, and a slot **134** into which a memory card can be inserted. Here, although examples of the connectors **132** include SDI, Gen-Lock, and Ethernet, the connectors **132** are not limited thereto and can be variously modified and changed within the scope of the gist thereof.

(15) An exterior of the image capturing apparatus **100** is provided with a first inlet port **151** and a second inlet port **161** for drawing in air from the outside of the image capturing apparatus **100**. Although, in the present exemplary embodiment, two inlet ports are provided, it is sufficient for at least one inlet port to be provided. The exterior of the image capturing apparatus **100** is provided with a first outlet port **135** and a second outlet port **136**.

(16) Outside air is introduced into the image capturing apparatus **100** from the first inlet port **151** and the second inlet port **161**, and is discharged from the first outlet port **135** to thereby make it possible to ventilate/cool the inside of the image capturing apparatus **100**.

(17) The image capturing apparatus **100** has threaded holes **171**, and the image capturing apparatus **100** can be fixed to, for example, a wall, a ceiling, a pan head, or a tripod.

(18) FIG. **4** is an exploded perspective view of the image capturing apparatus **100** of the present exemplary embodiment, the image capturing apparatus **100** being divided into 10 units. A front unit **110**, an intermediate unit **120**, a rear unit **130**, a top unit **140**, a right unit **150**, a left unit **160**, and a bottom unit **170** constitute a housing (exterior) of the image capturing apparatus **100**. A filter unit **199**, a sensor unit **180**, and a cooling unit **190** are provided inside the image capturing apparatus **100**.

(19) The front unit **110** constitutes a front surface of the housing of the image capturing apparatus **100**, and has the lens mount **111** and the lens connector **112**.

(20) The filter unit **199** has a mechanism and an actuator that drive, for example, an ND filter or an IR cut filter, and various filters can be inserted and removed by an operation of a user.

(21) The intermediate unit **120** constitutes an intermediate portion of the housing of the image capturing apparatus **100**, and has a structure in which each unit is attached thereto.

(22) The sensor unit **180** includes an image capturing element **181**, serving as a first heat source, and a heat sink **182**, serving as a first heat-dissipating member for dissipating heat generated by the image capturing element **181**. The heat sink **182** is thermally connected to the first heat source, and absorbs the heat of the first heat source and dissipates the heat to air inside the image capturing apparatus **100**.

(23) The cooling unit **190** has ducts **191** and **193** and a fan **192**. The ducts **191** and **193** are ducts for guiding air drawn in from the first inlet port **151** and the second inlet port toward the first outlet port **135**, and the heat sink **182** is disposed inside the duct **191**. The fan **192** is disposed between the

heat sink **182** and the first outlet port **135** inside the ducts **191** and **193**. The fan **192** is a fan for discharging air toward the first outlet port **135** from the first inlet port **151** and the second inlet port **161**. The fan **192** draws in outside air from the first inlet port and the second inlet port. Since the drawn-in air flows between and in the vicinity of heat-dissipating fins of the heat sink **182**, air subjected to heat dissipation by the heat sink **182** flows and is drawn in by the fan **192**. The air that has been drawn in by the fan **192** (that is, the air subjected to heat dissipation by the heat sink **182**) flows in the duct **193**, is discharged from the first outlet port **135**, and is thus discharged. In an assembled state of the image capturing apparatus **100**, the heat sink **182** is contained inside the duct **191**, and thus it is possible to forcefully air-cool the first heat source (the image capturing element **181**). The image capturing apparatus **100** also has a data processing substrate **194** on which a heating element (for example, a data processing engine or an FPGA (not shown)), serving as a second heat source, is mounted. Although, in the present exemplary embodiment, the fan **192** draws in outside air by using a centrifugal fan, for example, an axial fan can be used depending upon the design by a designer. Therefore, a fan of any other form may be used as long as the fan can introduce outside air into the image capturing apparatus **100**. Although, in the present exemplary embodiment, a second heat source is mounted on one data processing substrate, a plurality of second heat sources may be separately mounted on a plurality of substrates.

(24) The rear unit **130** constitutes a rear surface of the housing of the image capturing apparatus **100**, and has the aforementioned various connectors **132**, the aforementioned various keys **133**, and an interface substrate (not shown) on which the various connectors **132** and the various keys **133** are mounted. The rear unit **130** also has a heat-dissipating metal plate **137**, serving as a second heat-dissipating member, and the heat-dissipating metal plate **137** is thermally connected to the heating element mounted on the data processing substrate **194** through heat-dissipating sheets **138**. Examples of materials of the heat-dissipating metal plate **137** include metals having high thermal conductivity, such as an aluminum alloy and a copper alloy. The heat-dissipating metal plate **137** is a second heat-dissipating member for dissipating heat of the second heat source (the data processing substrate **194**). That is, similarly to the first heat-dissipating member, the second heat-dissipating member absorbs heat generated by the second heat source and dissipates the heat to air inside the image capturing apparatus **100**.

(25) For the top unit **140**, metals having high thermal conductivity, such as an aluminum alloy or a copper alloy, are used as primary materials. Since the top unit **140** and the heat-dissipating metal plate **137** are thermally connected through heat-dissipating sheets **141**, heat of the data processing substrate **194** is transferred to the top unit **140** and is diffused to the outside.

(26) The right unit **150** and the left unit **160** each have a corresponding one of the inlet port **151** and the outlet port **161**, and are connected to the ducts **191** and **193**. Similarly to the top unit **140**, metals having high thermal conductivity, such as an aluminum alloy or a copper alloy, are used as primary materials, and it is possible to diffuse heat of the data processing substrate **194** to the outside by thermally connecting the right unit **150** and the left unit **160** to the heat-dissipating metal plate **137** through heat-dissipating sheets **152**.

(27) The bottom unit **170** has the threaded holes **171** for fixing the image capturing apparatus **100**. Similarly to the top unit **140**, metals having high thermal conductivity, such as an aluminum alloy or a copper alloy, are used as primary materials, and it is possible to diffuse heat of the data processing substrate **194** to the outside by thermally connecting the bottom unit **170** to the heat-dissipating metal plate **137** through heat-dissipating sheets **172**.

(28) As described above, the second heat-dissipating member (the heat-dissipating metal plate **137**) is connected to at least one unit constituting the exterior (the housing) of the image capturing apparatus **100** so as to transfer heat of the second heat-dissipating member to the at least one unit. This makes it possible to cool the second heat source (the data processing substrate **194**). This is realized in combination with a cooling method described below.

(29) FIG. 5 is a sectional view of the image capturing apparatus **100** after removal of the front unit

110 and the filter unit **199**. Detailed structures of the sensor unit **180** and the cooling unit **190** and a cooling structure of the image capturing apparatus **100** are described below with reference to FIG. 5. Outer surfaces of the sensor unit **180** are constituted by a front sensor case **183** and a rear sensor case **184**, and the image capturing element **181** is disposed inside the sensor unit **180**. The image capturing element **181** is thermally connected to a heat-absorption surface of a Peltier element **14** through a heat conduction member **11**, a metal block **13**, and a heat conduction member **12**. The so-called heat conduction members **11** and **12** may be, for example, heat-dissipating sheets or heat-dissipating grease. The metal block **13** is desirably made of a material having high thermal conductivity, such as an aluminum alloy or a copper alloy. A heat-dissipating surface of the Peltier element **14** is thermally connected to the heat sink **182** through a heat conduction member (for example, a heat-dissipating sheet or heat-dissipating grease) and a rear sensor case **184**. It is possible to cool the image capturing element **181** by dissipating heat of the heat sink **182** to air inside the image capturing apparatus **100**. Here, the rear sensor case **184** is also desirably made of a material having high thermal conductivity, such as an aluminum alloy or a copper alloy.

(30) A case sealing member **185** is provided between the front sensor case **183** and the rear sensor case **184**, and prevents entry of water vapor into the sensor unit **180** and condensation in the image capturing element **181**. The front sensor case **183** has an open window provided in accordance with the position of an image capturing surface of the image capturing element **181**, and a cover glass **187** is attached to the front sensor case **183** through a glass sealing member **186**. Although, in the present exemplary embodiment, the image capturing element **181** is cooled through the Peltier element **14**, the image capturing element **181** may be directly connected to the heat sink **182** without using a Peltier element.

(31) An air path A (a first air path) is formed such that outside air is drawn in through the duct **191** from the inlet port **151** and the inlet port **161** when the fan **192** attached to the cooling unit **190** is driven. The air path A is a path through which a current of air for guiding air drawn in from the inlet port **151** and the inlet port **161** up to the fan **192** passes, and is formed by the duct **191**. At this time, since the heat sink **182** is contained in the duct **191** (the first heat-dissipating member is disposed at the air path A), it is possible to cool the heat-dissipation surface of the Peltier element **14** and the image capturing element **181**. A duct **198** and an air path B (a second air path) are formed by the duct **193**, a duct cover **195**, and a duct **196**. The air path B is a path through which a current of air for guiding air discharged from the fan **192** up to the first outlet port **135** passes. The air discharged from the fan **192** passes through the air path B and is discharged to the outside of the image capturing apparatus **100** from the first outlet port **135**. That is, the air paths A and B are paths through which air passes and are formed by the duct **198**, and the inlet port **151** and the inlet port **161** are at a most upstream side of the duct **198** and the first outlet port is at a most downstream side of the duct **198**.

(32) FIG. 6 is a perspective view of the image capturing apparatus **100** after taking out only the cooling unit **190** and the duct **196**. Details of a cooling mechanism of the data processing substrate **194** is described with reference to FIGS. 5 and 6.

(33) As shown in FIG. 6, the duct **193** has a plurality of openings **197**. The openings **197** are positioned between the fan **192** and the second outlet port. Although, in the present exemplary embodiment, a plurality of openings are provided, one opening in which a plurality of holes are connected to each other may be provided. That is, it is sufficient to provide at least one opening in the duct **193**. By providing a structure such as that described above, part of air discharged toward the first outlet port from the fan **192**, that is, part of air discharged by passing through the duct **198** (air flowing through the air path B) is pushed out from the openings **197**.

(34) As shown in FIG. 5, an air path C (a third air path) is formed such that the air pushed out from the openings **197** circulates inside the image capturing apparatus **100**, and is discharged to the outside of the image capturing apparatus **100** from the second outlet port **136**. The air path C is not formed by a duct as with the air path A and the air path B. Specifically, the air path C is formed by

at least one unit constituting the exterior of the image capturing apparatus **100**, such as the rear unit **130** or the top unit **140**. This is to cool a heat source (such as a power supply circuit substrate) other than the data processing substrate **194** by ventilation inside the image capturing apparatus **100**. In other words, in order to cool a plurality of heat-dissipating members for dissipating heat of a plurality of second heat sources, the plurality of second heat-dissipating members are disposed at the air path C. It is sufficient for at least one second heat-dissipating member for dissipating heat of at least one second heat source to be disposed at the air path C. The openings **197** are at a most upstream side of the air path C, and the second outlet port is at a most downstream side of the air path C. Here, by disposing the second heat-dissipating member (the heat-dissipating metal plate **137**) between the openings **197** and the second outlet port **136** (disposing the second heat-dissipating member at the air path C), air passing through the air path C is supplied to the heat-dissipating metal plate **137**. Therefore, the air that is pushed out from the openings **197** cools the heat-dissipating metal plate **137** and is discharged from the second outlet port. As described above, if a plurality of second heat-dissipating members are disposed, these can also be cooled. Although, in the present exemplary embodiment, the second outlet port **136** opens in the rear surface of the housing of the image capturing apparatus **100**, for example, as shown in FIG. 7, a widened clearance of the connectors **132** and the housing may be the second outlet port **136**. That is, the second outlet port may be smaller than the first outlet port.

(35) Here, the temperature of air at an inlet of the air path A, that is, the temperature of air that has been drawn in from the first inlet port **151** and the second inlet port **161** is substantially the same as the temperature of the environment in which the image capturing apparatus **100** is installed. The temperature of air in the air path B increases by heat dissipated from the heat sink **182**. As described above, part of the air in the air path B is pushed out from the openings **197** and flows to the air path C, and the remaining air pushed out from the openings **197** is discharged from the first outlet port **135**.

(36) In other words, part of air heated by the heat sink **182** flows to the air path C. Since the second heat-dissipating member is disposed at the air path C, in order to cool the second heat-dissipating member (the heat-dissipating metal plate **137**), it is sufficient that the condition of (temperature of air in the air path B) < (temperature of the heat-dissipating metal plate **137**) be satisfied. As long as the condition above is satisfied, the data processing substrate **194** is cooled by the air pushed out to the air path C. Therefore, it is desirable that the temperatures of the respective portions in the cooling unit **190** have the following relationship:

(environment temperature) < (temperature of air in the air path B) < (temperature of the heat-dissipating metal plate **137**)

(37) In general, as the temperature of the image capturing element **181** decreases, the performance of the image capturing element **181** is improved, whereas the heat value itself of the image capturing element is lower than that of, for example, a data processing engine. Therefore, if an image capturing apparatus has a structure such as that of the present exemplary embodiment, the fan **192** is driven so as to reduce the temperature of the heat sink **182** to a temperature near the environment temperature and thus causes the temperature of air that flows in the air path B or the temperature of air at the inlet of the air path C to be a temperature near the environment temperature. On the other hand, elements such as a data processing engine and an FPGA, have heat values larger than those of, for example, image capturing elements, whereas the operations of such elements can often be guaranteed even if their temperatures are rather high. Therefore, if an image capturing apparatus has a structure such as that of the present exemplary embodiment, in general, the temperature of the heat-dissipating metal plate **137** is increased, and the relationship of (temperature of air in the air path B) < (temperature of the heat-dissipating metal plate **137**) mentioned above is established.

(38) When the structure of the image capturing apparatus **100** according to the present exemplary embodiment is used, even if a plurality of heat sources are mounted so as to be scattered on the data

processing substrate **194**, the heat sources can be cooled by air flow in the air path C without changing the shape of the duct **198** and ventilation resistance. As shown in FIG. **8**, a plurality of heat-dissipating metal plates or a plurality of heating elements, serving as second heat sources, may be scattered on a plurality of substrates. The heat source of the data processing substrate **194** is such that heat is diffused to, in addition to the heat-dissipating metal plate **137**, a second heat-dissipating metal plate **237** through a heat-dissipating sheet **232**. A heat source (not shown) on a power supply circuit substrate **231** is such that heat is diffused to a third heat-dissipating metal plate **233** through a heat-dissipating sheet **232**. Even in these cases, by providing openings **297A**, **297B**, and **297C** in accordance with the heat-dissipating metal plates **237**, **137**, and **233**, it becomes possible to efficiently cool a plurality of heat sources and to improve the cooling performance inside the image capturing apparatus **100**.

(39) The duct **193** is disposed below the second heat source (the data processing substrate **194**), and the openings **197** are provided in an upper surface of the duct **193**. Therefore, when dirt of outside air, such as dust, is drawn in by the fan **192**, the dirt does not flow to, for example, the data processing substrate **194** and falls due to gravitation. The dirt passes through the air path B and is discharged to the outside of the image capturing apparatus **100** from the first outlet port **135**.

Second Exemplary Embodiment

(40) An outdoor installation-type image capturing apparatus according to an exemplary embodiment is described below. FIG. **9** is a perspective view of an image capturing apparatus **300** according to the present exemplary embodiment, and FIG. **10** shows an internal structure of the image capturing apparatus **300**.

(41) The image capturing apparatus **300** includes a housing **310**, and a camera device **320** and a lens **330** that are contained in the housing **310**. Here, the housing **310**, the camera device **320**, and the lens **330** may each be removably provided or may be integrated with each other.

(42) Here, the camera device **320** has a structure that is the same as the structure of the image capturing apparatus **100** described in the first exemplary embodiment. By driving a fan (not shown) inside the camera device **320**, air is drawn in from an inlet port **321** and is discharged from a first outlet port **135** and a second outlet port **136**.

(43) The housing **310** is hermetically sealed and causes the camera device **320** and the lens **330** to be waterproof. Therefore, the fan of the camera device **320** causes air inside the housing **310** to circulate. Even in this case, similarly to the image capturing apparatus **100** described in the first exemplary embodiment, it is possible to efficiently cool heat sources, such as an image capturing element and a data processing engine.

(44) Although exemplary embodiments of the present invention have been described above, the present invention is not limited to these exemplary embodiment, and various modifications and changes are possible with the scope of the spirit of the present invention.

(45) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(46) This application claims the benefit of Japanese Patent Application No. 2022-102256, filed Jun. 24, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image capturing apparatus whose exterior is provided with at least one inlet port for drawing in air from outside of the image capturing apparatus and a first outlet port and a second outlet port for discharging air from inside of the image capturing apparatus, the image capturing apparatus comprising: a first heat-dissipating member configured to dissipate heat generated by a first heat source; at least one second heat-dissipating member configured to dissipate heat generated by at

least one second heat source; a duct inside which the first heat-dissipating member is disposed, and configured to guide air drawn in from the at least one inlet port toward the first outlet port; and a fan disposed between the first outlet port and the first heat-dissipating member inside the duct and configured to discharge air toward the first outlet port from the at least one inlet port, wherein at least one opening is provided in the duct, wherein the at least one opening is positioned between the fan and the first outlet port, wherein part of the air discharged toward the first outlet port by the fan is pushed out from the at least one opening, wherein the air pushed out from the at least one opening cools the at least one second heat-dissipating member and is discharged from the second outlet port, and wherein at least one unit constituting the exterior of the image capturing apparatus forms a third air path for guiding the air pushed out from the at least one opening to the second outlet port, and the at least one second heat-dissipating member is disposed at the third air path.

2. The image capturing apparatus according to claim 1, wherein the first heat source is an image capturing element, and the at least one second heat source is a data processing substrate.
3. The image capturing apparatus according to claim 1, wherein the first heat-dissipating member is a heat sink, and the at least one second heat-dissipating member is a heat-dissipating metal plate.
4. The image capturing apparatus according to claim 1, wherein the duct is disposed below the at least one second heat source, and the at least one opening is provided such that the air pushed out from the at least one opening flows to a position of the at least one second heat-dissipating member.
5. The image capturing apparatus according to claim 1, wherein the at least one second heat-dissipating member is connected to at least one unit constituting the exterior of the image capturing apparatus so as to transfer heat to the at least one unit.
6. The image capturing apparatus according to claim 1, wherein the duct forms a first air path for guiding the air drawn in from the at least one inlet port up to the fan, and the at least one second heat-dissipating member is disposed at the first air path.
7. The image capturing apparatus according to claim 1, wherein the duct forms a second air path for guiding the air discharged from the fan up to the first outlet port, and the at least one opening is provided in the duct such that part of air that flows in the second air path is pushed out from the at least one opening.
8. The image capturing apparatus according to claim 1, wherein a temperature of the at least one second heat-dissipating member is higher than a temperature of air that flows in a second air path.
9. The image capturing apparatus according to claim 1, wherein the second outlet port is smaller than the first outlet port.
10. The image capturing apparatus according to claim 1, wherein the second outlet port is provided at a connector configured to allow the image capturing apparatus to send a signal to and receive a signal from an external device.
